

Yash Bhavsar

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EDUCATION

McMaster University

June 2025

Electrical Engineering (Co-op) B.A.Sc.

Hamilton, ON

Relevant Courses: Object-Oriented Programming(OOP), Data Structures and Algorithms, Fundamentals of Machine Learning, Advanced Internet Communications, Computer Architecture, Microprocessor Systems

EXPERIENCE

Finaptive Solutions, Software Engineer

Feb 2025 – Current

- **Sole engineer** at a 15-person financial analytics startup, owning full software architecture, backend infrastructure, and compliance tooling while collaborating directly with leadership and analysts
- Built asynchronous report-processing pipeline with **Flask, Celery, Redis, PostgreSQL**, cutting generation time from 40+ minutes to **under 5 minutes**
- Achieved **SOC1 and GDPR** compliance independently by implementing AES256 encryption via Azure Key Vault, complete audit logging, and role-based access controls
- Led migration of an ERP data analytics platform from PHP/Symfony 3.2 to a modern **Azure**-based stack, resolving **50+** production issues and integrating secure, automated data pipelines between systems
- Designed and deployed an admin portal with dynamic override controls, audit log exports, user provisioning, and secure password generation, improving internal workflows by **60%**

AMD, Software Engineer Intern

May 2023 – Apr 2024

- Led LINT-based **code review across 10+ long-term projects**, addressing 100+ critical errors while coordinating with many cross-functional engineering teams and ensuring clear and maintainable code
- Developed and deployed automated tests using **C Shell** and **Python**, streamlining code design and testing, resulting in **20+ hours saved** per week
- Implemented and maintained a dashboard utilizing **PHP, JavaScript**, and **Perl**, to present real-time test results fetched from a **MySQL** database, reducing the weekly workload by removing the need for manual updates

PROJECTS

Lunar Lander Reinforcement Learning Hackathon (1st Place)

- Developed **Deep Q-Networks (DQN)** and **Proximal Policy Optimization (PPO)** agents using **PyTorch**, achieving a winning average test reward of **285.30**
- Optimized performance with **hyperparameter tuning**, experience replay, and trust region updates, improving performance by **29.8%**

Intelligent Email Sorting Chrome Extension

- Achieved 90% email classification accuracy with **spaCy** and **NLTK** models for sorting (work, personal, urgent, promotional)
- Built **Node.JS** backend with real-time email retrieval via **Gmail, Outlook, and Yahoo APIs**
- Reduced **inbox management time by 30%** through extensive testing and **GDPR-compliant** optimizations

Embedded LiDAR Mapper

- Engineered an embedded **C** program for a TI micro-controller, using I2C communication, sensor initialization, precise distance measurements, and motor control to parse raw Li-DAR data
- Created a **Python** script for organizing and visualizing data from micro-controller through UART communication and matplotlib, including data storage, organization options, and 3D scatter plot visualization

TECHNICAL SKILLS

Programming: 5+ years of experience including Python, C/C++, JavaScript, Java, PHP, SQL, HTML5/CSS3, Flask, Node.js, Express.js, React.js, Celery, Redis, PyTorch

Databases and Cloud: PostgreSQL, MongoDB, MySQL, Azure, Docker, Linux, GitHub

Security and Testing: AES256 encryption, RBAC, Audit Logging, GDPR, SOC1, Unit Testing, LINT, CI/CD